AQUAREV WATER · COMMERCIAL POOL SYSTEMS · DRAWING SET 100

# AquaRev Water Conditioning Device Reference Installation Drawings

Sample commercial pool and spa pump room, for engineers new to the device · in-line on the return, downstream of all treatment equipment

## SHEET INDEX


| | |
|---|---|
| G-001 | Cover sheet, general notes, abbreviations & legend |
| C-101 | Site plan · pool, spa, equipment room, limit of work |
| M-101 | Pump room equipment plan · 1/2" = 1'-0" |
| M-201 | Recirculation flow diagram · pool and spa loops |
| M-301 | Device & connection details · dimensions, union joint, cut length, support |
| M-302 | Installation sequence · eight steps, lock-out to restart |
| M-401 | Orientation & alternate arrangements · vertical, offset, parallel |
| M-501 | Materials & records · materials, tools, installation record template, contacts |


## DESIGN BASIS · SAMPLE PROPERTY USED ON EVERY SHEET

| LOOP | VOLUME | TURNOVER | FLOW | RETURN | DEVICE |
|---|---|---|---|---|---|
| Pool · WCD-1 | 150,000 gal | 6.0 h | 417 gpm | 4" PVC | KD4X8X4 |
| Spa · WCD-2 | 3,000 gal | 0.5 h | 100 gpm | 2" PVC | KD2X4X2 |

Model ranges: KD4X8X4 240–480 gpm · KD2X4X2 44–127 gpm · rated head loss 2.6 psi each. Source: Product Specification Guide 2026.

## REFERENCE DOCUMENTS & STANDARDS

IAPMO R&T Product Listing, File No. W-12547 · NSF/ANSI/CAN 50-2024 · USPC® mark
Listed in compliance with the Uniform Swimming Pool, Spa and Hot Tub Code
AquaRev Product Specification Guide 2026 · Pump Room Implementation Guide Rev 1.1
ASTM D2564 (PVC solvent cement) · ASTM F656 (primer) · ASTM D2855 (solvent-cemented joints)
Local building and public pool health codes · AHJ written permit determination on file

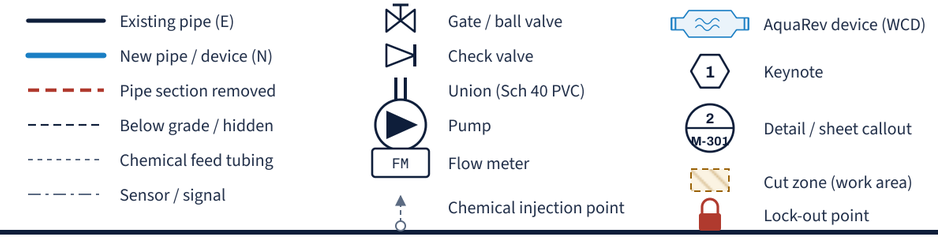
## ABBREVIATIONS

| | | | |
|---|---|---|---|
| AHJ | Authority having jurisdiction | GV | Gate or ball valve |
| BW | Backwash | LOTO | Lock-out, tag-out |
| CP | Chemical controller | MP | Chemical metering pump |
| CV | Check valve | MV | Multiport (backwash) valve |
| DS | Disconnect switch | N.I.C. | Not in contract |
| (E) | Existing, to remain | OD | Outside diameter |
| (N) | New | SCH | Schedule (pipe wall) |
| FD | Floor drain | TYP | Typical |
| FM | Flow meter | WCD | Water conditioning device (AquaRev) |

## GENERAL NOTES

1. This is a reference set drawn around a sample property. It is not issued per site; the same set serves every installation. Verify existing conditions, pipe sizes, flow and clearances in the field before ordering or cutting.
2. The AquaRev device is a passive in-line water conditioning device certified to NSF/ANSI/CAN 50-2024 (IAPMO R&T File No. W-12547). It supplements the pool's disinfection system and does not replace it. Sanitizer and pH are maintained within the regulated range at all times, exactly as before.
3. Install one device per recirculating loop, on the return line, downstream of the pump, filter, heater, chemical feeders and sensors, as close to the pool return as practical. Any orientation is acceptable.
4. Existing equipment is not modified. No electrical, gas, water-supply, drain or sewer connection is made. Suction-side piping, main drains and drain covers are not touched.
5. Size by return pipe diameter, then confirm the loop flow falls inside the model range (1/M-301). Refer flow within 10% of a range limit, pools over 500,000 gal, and 8 in. or larger returns to AquaRev before ordering. Loops above a model's maximum may be split across two devices in parallel (M-401).
6. The device adds up to 2.6 psi at rated flow. Confirm pump margin with a clean filter before installing.
7. Connections: two Schedule 40 PVC unions supplied with the device, solvent-welded to the existing pipe. Primer ASTM F656, cement ASTM D2564, practice ASTM D2855. Heavy-body cement for 4 in. and larger.
8. No dedicated supports are required. Existing pipe supports remain; on a long unsupported run, support near the device (4/M-301).
9. Lock out and tag the pump disconnect and every chemical feeder on the loop before opening the line. Depressurize and verify the filter gauge reads zero before cutting.
10. Allow the cement manufacturer's full cure for the pipe size and temperature before pressurizing.
11. Work is performed by a licensed pool contractor or plumber. The warranty requires professional installation.
12. Obtain the AHJ's written permit determination before install day (Implementation Guide §6–7). The contractor carries a copy on site.
13. Record model, serial, cut length, cure time and gauge readings on the installation record (M-501).
14. Where these drawings differ from the union or cement manufacturer's instructions, the manufacturer's instructions govern.

## LEGEND




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PROJECT
**Sample Commercial Pool Pump Room**
AquaRev water conditioning device installation

STATUS
Reference set. Not site-specific.

FOR
Engineers new to the device

SHEET TITLE
**Cover Sheet, General Notes & Legend**

| | |
|---|---|
| DRAWN | AquaRev Water |
| BASIS | Sample property |
| DATE | 22 SEP 2026 |
| SCALE | As noted |
| SIZE | ANSI B 17 × 11 |
| SET | 100 · REV A |

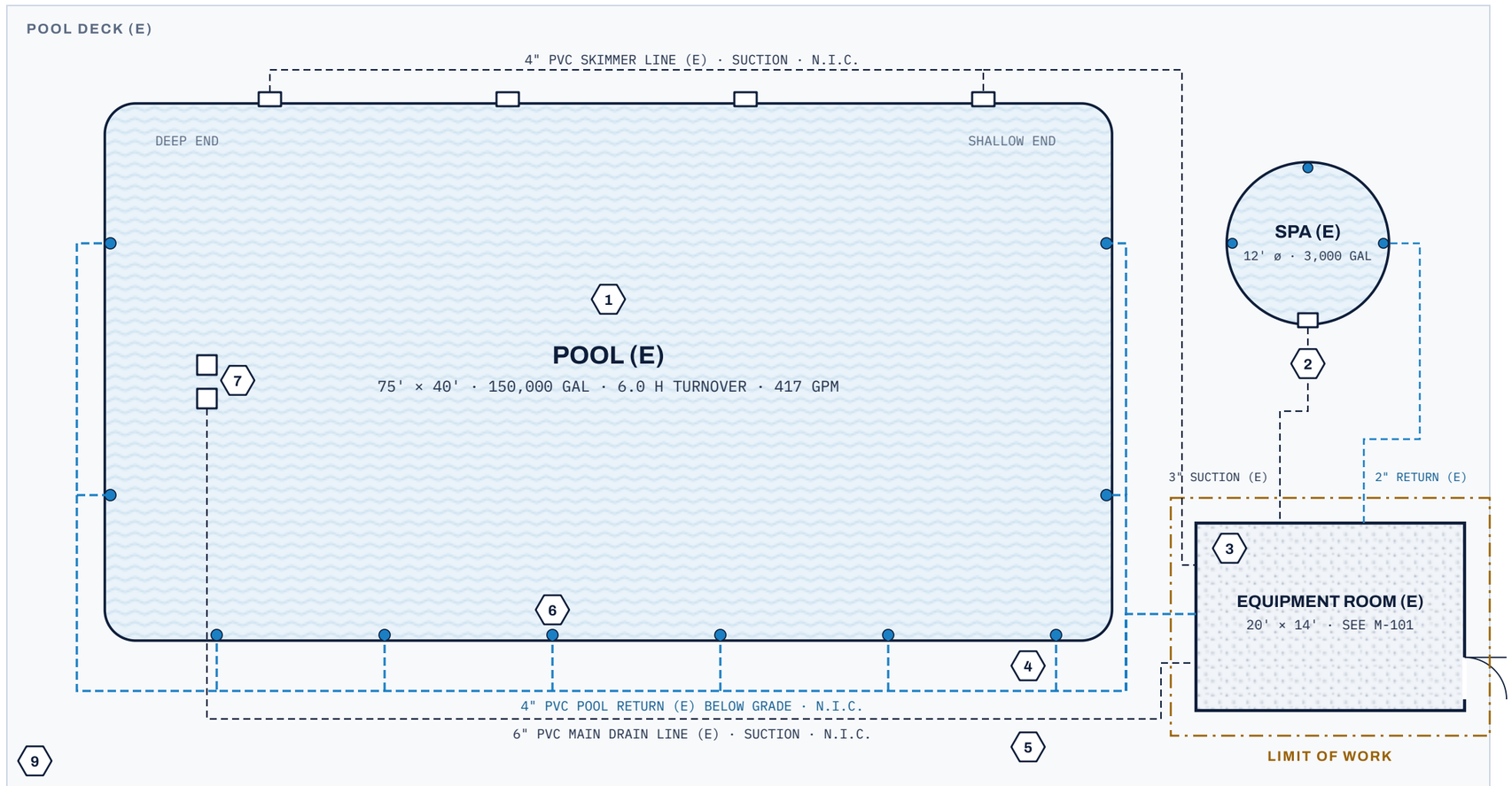
REVISIONS

| REV | DATE | DESCRIPTION |
|---|---|---|
| A | 09-22-26 | Reference issue |

Sample arrangement. Field-verify all existing pipe sizes, materials, flow rates and clearances before ordering or cutting. Existing equipment shown is not modified.

REFERENCE SET. NOT A SITE-SPECIFIC DRAWING.

SHEET
**G-001**
SHEET 1 OF 8




## SITE DATA · SAMPLE PROPERTY

| | |
|---|---|
| Bodies of water | 2 (pool, spa) · one loop each |
| Pool volume / turnover | 150,000 gal · 6.0 h · 417 gpm |
| Spa volume / turnover | 3,000 gal · 0.5 h · 100 gpm |
| Return pipe, pool / spa | 4" PVC · 2" PVC (field verify OD) |
| Equipment room | 20' × 14' at grade, door to service path |
| Devices | WCD-1 KD4X8X4 · WCD-2 KD2X4X2 |
| Pool closure | 3 to 5 h per loop incl. cement cure |

1 C-101 **SITE PLAN · TYPICAL**

1" = 10'-0" · WORK CONFINED TO THE EQUIPMENT ROOM

SCALE 1" = 10'-0" (AT 17 × 11 IN.)

0 10' 20' 30'

## KEYNOTES

1. Existing pool, 150,000 gal. Water, shell, deck and barriers not touched.
2. Existing spa, 3,000 gal, separate recirculation loop. Gets its own device (WCD-2).
3. Equipment room. All work occurs here. Enlarged plan on M-101.
4. Existing 4" pool return, below grade to wall inlets. Not modified outside the room.
5. Existing suction lines (main drain, skimmers, spa). Suction side is not touched.
6. Existing return inlets, N.I.C.
7. Existing main drains and covers, N.I.C. No suction-side or barrier work is involved.
8. Service access. Device cartons (up to 44" × 16.5" × 16.5", 50 lb) arrive here.
9. Pool closure signage and barrier during the 3 to 5 hour closure window, per property policy.

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PROJECT

**Sample Commercial Pool Pump Room**

AquaRev water conditioning device installation

STATUS

Reference set. Not site-specific.

FOR

Engineers new to the device

SHEET TITLE

**Site Plan**

| | |
|---|---|
| DRAWN | AquaRev Water |
| BASIS | Sample property |
| DATE | 22 SEP 2026 |
| SCALE | 1" = 10'-0" |
| SIZE | ANSI B 17 × 11 |
| SET | 100 · REV A |

REVISIONS

| REV | DATE | DESCRIPTION |
|---|---|---|
| A | 09-22-26 | Reference issue |

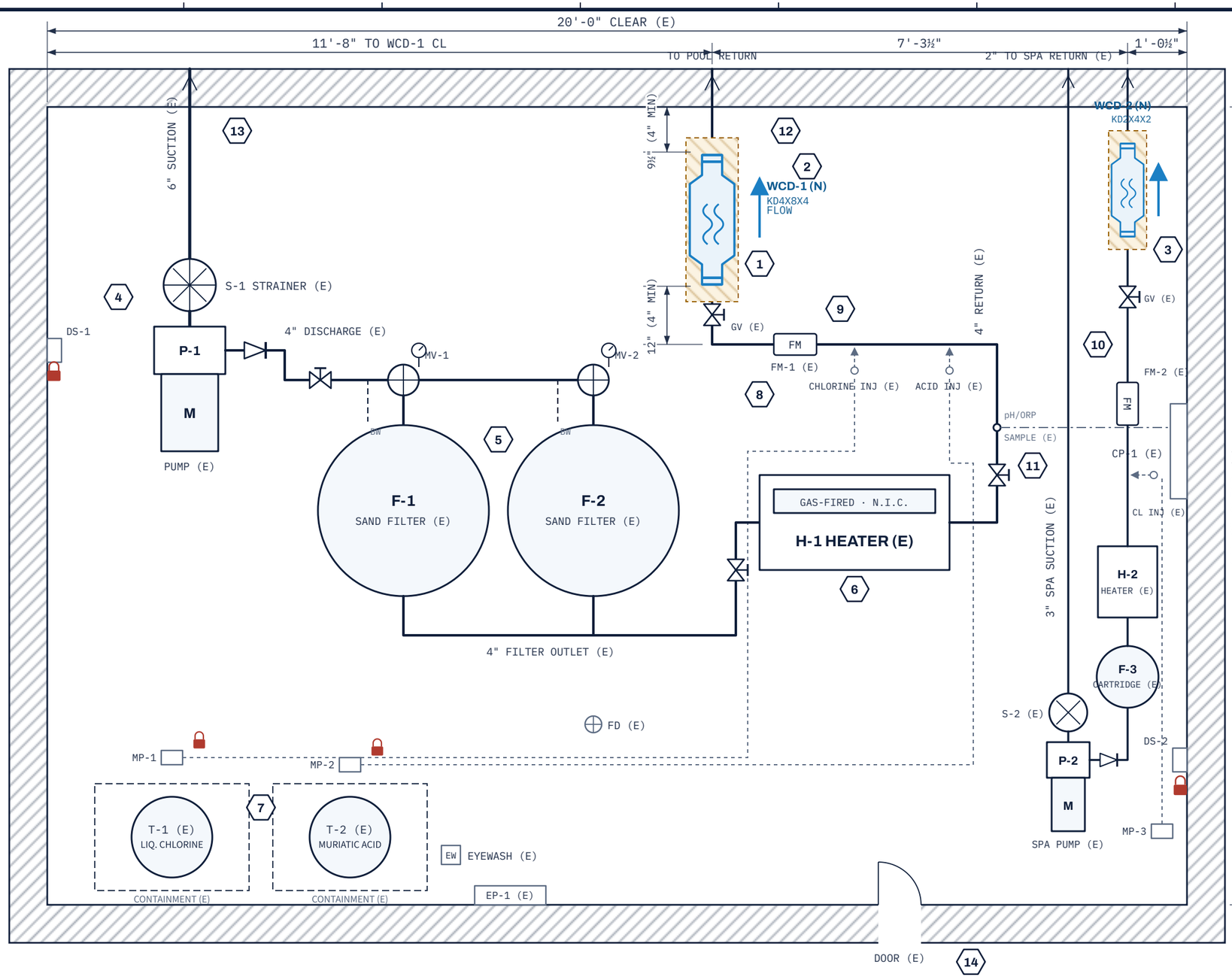
Sample site, drawn to show where the work sits on a property. Buried pipe routes are diagrammatic; no excavation is part of this work.

REFERENCE SET. NOT A SITE-SPECIFIC DRAWING.

SHEET

**C-101**

SHEET 2 OF 8




**1 / M-101** **PUMP ROOM EQUIPMENT PLAN · TYPICAL**

1/2" = 1'-0" · TWO LOOPS, ONE DEVICE EACH · DEVICE IS THE LAST COMPONENT BEFORE THE RETURN

GRAPHIC SCALE · 1/2" = 1'-0" AT 17 × 11 IN.

0 1' 2' 4' 6'

## KEYNOTES

1. Existing 4" PVC return. Remove one section, weld union tails. 3/M-301.
2. WCD-1 (N) KD4X8X4 in 4" unions, flow arrow toward pool. 1/M-301, 2/M-301.
3. WCD-2 (N) KD2X4X2 on 2" spa return, same detail at 2" size.
4. Existing pump and strainer, N.I.C. Lock out at DS-1 / DS-2 before work.
5. Existing filters and multiport valves, N.I.C. Backwash piping not shown.
6. Existing heater, N.I.C. Device is downstream of heater outlet.
7. Existing chemical storage, metering pumps and tubing, N.I.C. Lock out MPs.
8. Existing flow meter. Record reading before and after installation.
9. Existing return isolation valve where fitted; otherwise drain filter to isolate.
10. Existing injection points stay upstream of the device. No relocation.
11. Existing controller and sample line, N.I.C. Do not change setpoints.
12. Existing pipe supports remain. No dedicated support is added. 4/M-301.
13. Suction-side piping. Not touched.
14. Door. Open for ventilation while priming and cementing.

## PLAN NOTES

A. Clearance at each device: straight run of L1 + 8" minimum, D1 + 4" around the body. See 1/M-301 and E/M-401.

B. Cut length is field-measured with the union tails dry-fitted. Never cut to L1.

C. Where the return run is shorter than L1 + 8", reroute with elbows. M-401.

D. Loops with flow above one model's maximum: parallel devices. M-401.

E. Existing pipe drawn black; new work blue; cut zone hatched amber.

F. Existing pipe sizes are the design basis. Field-verify outside diameter.

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PROJECT
**Sample Commercial Pool Pump Room**
AquaRev water conditioning device installation

STATUS
Reference set. Not site-specific.

FOR
Engineers new to the device

SHEET TITLE
**Pump Room Equipment Plan**

| | |
|---|---|
| DRAWN | AquaRev Water |
| BASIS | Sample property |
| DATE | 22 SEP 2026 |
| SCALE | 1/2" = 1'-0" |
| SIZE | ANSI B 17 × 11 |
| SET | 100 · REV A |

REVISIONS

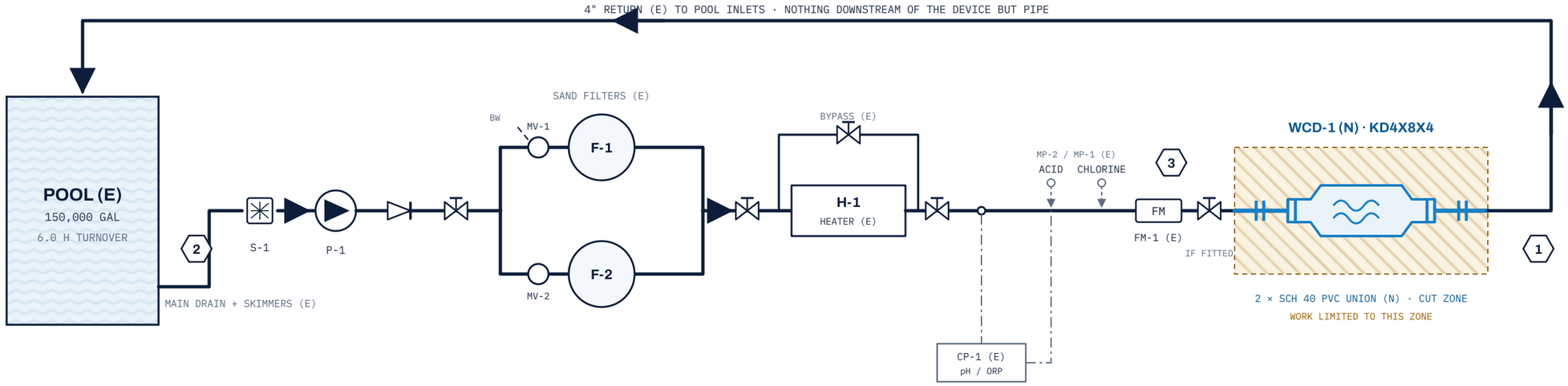
| REV | DATE | DESCRIPTION |
|---|---|---|
| A | 09-22-26 | Reference issue |

Equipment positions are typical. Pipe sizes shown are the design basis; measure outside diameter on site. Existing equipment (E) is N.I.C. and is not modified.

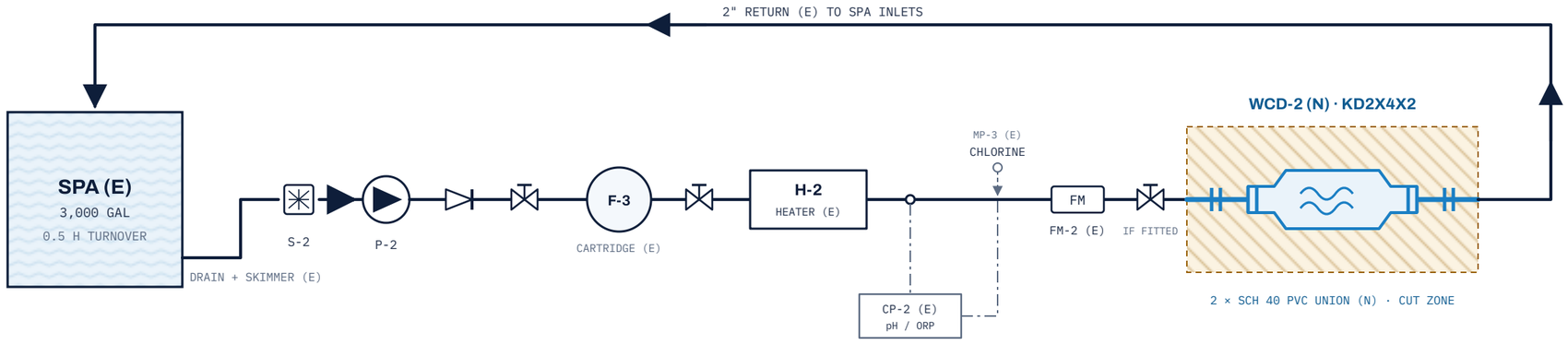
REFERENCE SET. NOT A SITE-SPECIFIC DRAWING.

SHEET
**M-101**
SHEET 3 OF 8

## POOL LOOP · 417 GPM · 4" PVC · ONE DEVICE (WCD-1)




## SPA LOOP · 100 GPM · 2" PVC · ONE DEVICE (WCD-2)




### SYMBOLS

- Hair and lint strainer (E)
- Circulation pump (E)
- Check valve (E)
- Isolation valve (E)
- FM — Flow meter (E)
- Chemical injection point (E)
- Sensor sample tap, signal (E)
- Union, Sch 40 PVC, supplied (N)
- AquaRev device (N)
- Cut zone: the only place pipe is opened

### KEYNOTES

1. Nothing is installed downstream of the device except existing pipe and inlets.
2. Suction side, main drains, covers and skimmers are not part of this work.
3. Sensors and injection points remain upstream of the device. Setpoints unchanged.

### HYDRAULIC CHECK (FILL BEFORE ORDERING)

| ITEM | POOL · WCD-1 | SPA · WCD-2 | SOURCE |
|---|---|---|---|
| Design flow (must fall inside model range) | 417 gpm · 240–480 | 100 gpm · 44–127 | Flow meter / pump curve; Spec Guide |
| Rated head loss added by device | ≤ 2.6 psi | ≤ 2.6 psi | Spec Guide 2026; listing footnote |
| Filter gauge, clean filter, before / after | _____ / _____ psi | _____ / _____ psi | Field reading, installation record |

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PROJECT
**Sample Commercial Pool Pump Room**
AquaRev water conditioning device installation

STATUS
Reference set. Not site-specific.

FOR
Engineers new to the device

SHEET TITLE
**Recirculation Flow Diagram**

| | |
|---|---|
| DRAWN | AquaRev Water |
| BASIS | Sample property |
| DATE | 22 SEP 2026 |
| SCALE | None (schematic) |
| SIZE | ANSI B 17 × 11 |
| SET | 100 · REV A |

REVISIONS

| REV | DATE | DESCRIPTION |
|---|---|---|
| A | 09-22-26 | Reference issue |

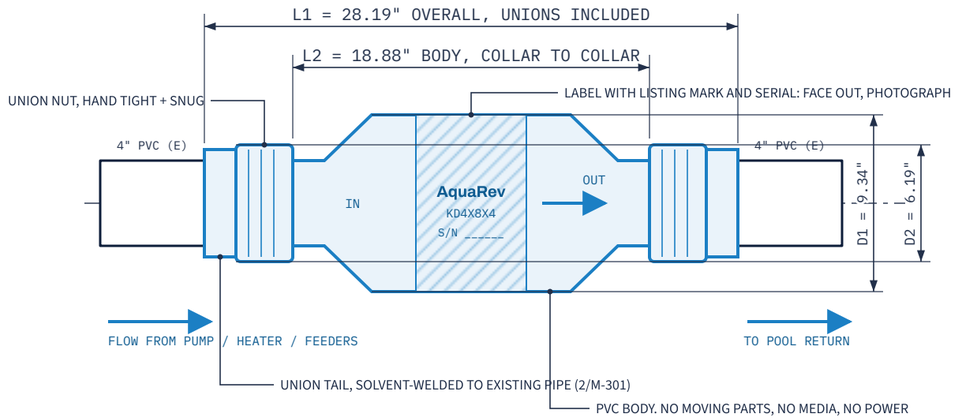
Single-line schematic, not to scale. Valves and bypasses are typical; the existing arrangement governs. The device is the last component before the return.

REFERENCE SET. NOT A SITE-SPECIFIC DRAWING.

SHEET
**M-201**
SHEET 4 OF 8

## 1 M-301 DEVICE ELEVATION · KD4X8X4 SHOWN

1½" = 1'-0" · ANY ORIENTATION · IN ARROW TOWARD PUMP, OUT TOWARD POOL

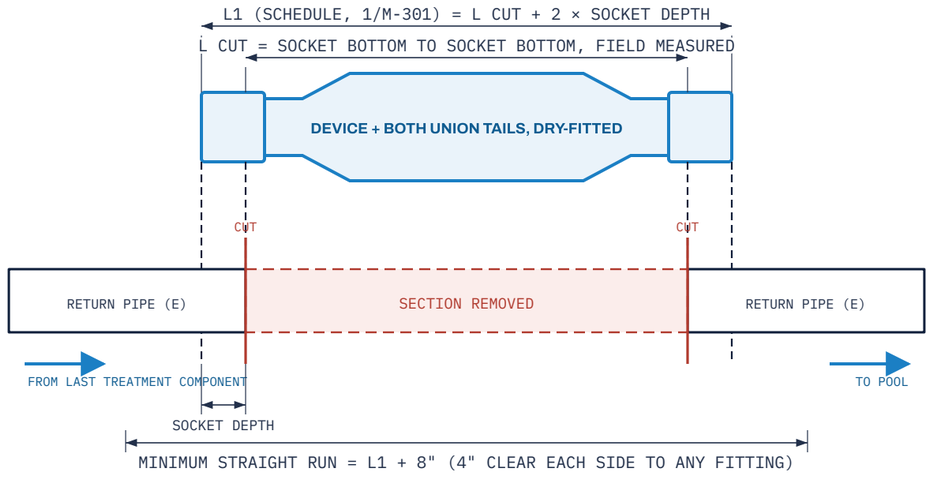



| MODEL | PIPE | L1 | L2 | D1 | D2 | FLOW GPM | SHIP LB |
|---|---|---|---|---|---|---|---|
| KD2X4X2 | 2" | 20.25" | 13.375" | 5.00" | 3.60" | 44–127 | 8 |
| KD3X6X3 | 3" | 24.00" | 15.00" | 7.25" | 5.06" | 112–273 | 20 |
| KD4X8X4 | 4" | 28.19" | 18.88" | 9.34" | 6.19" | 240–480 | 35 |
| KD6X12X6 | 6" | 40.00" | 25.00" | 13.75" | 9.00" | 440–1,100 | 50 |

HEAD LOSS 2.6 PSI ALL MODELS · 8" AND 10" ON REQUEST · SOURCE: PRODUCT SPECIFICATION GUIDE 2026

## 3 M-301 CUT LENGTH · FIELD MEASURE

NO SCALE · APPLIES TO EVERY MODEL

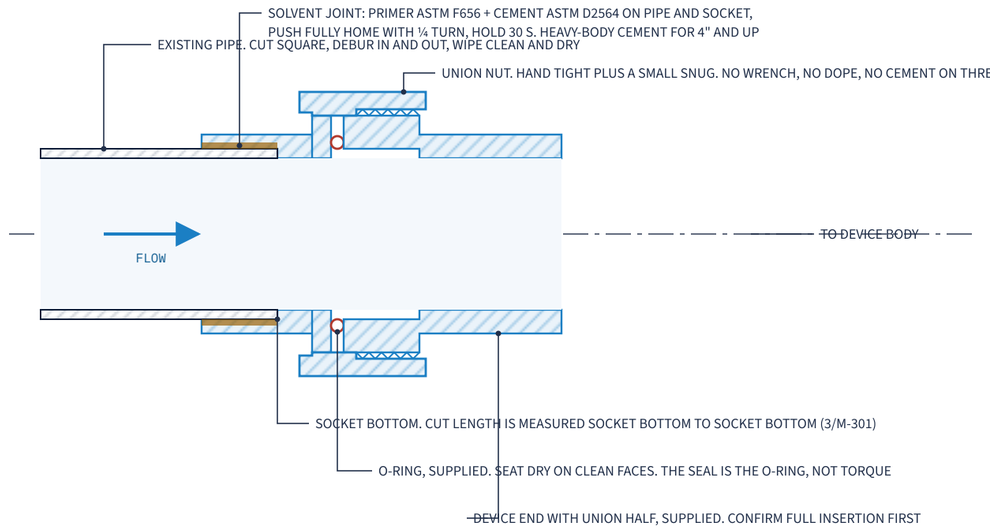



### NOTES

1. Never cut to L1. Dry-fit both union tails to the device, measure socket bottom to socket bottom, and mark the pipe from that.
2. Measure twice. A short cut is recovered with a coupling and spare pipe; a long cut is not.
3. Two-inch field example: L1 20.25 in., cut 17.5 in. Larger sizes have deeper sockets; measure the unit in hand.
4. Cut square. A square, deburred end seats fully in the socket and lets the union faces meet flat.
5. Locate the cut so the device sits after the last treatment component, as close to the pool return as the room allows.

## 2 M-301 UNION JOINT SECTION · TYPICAL BOTH ENDS

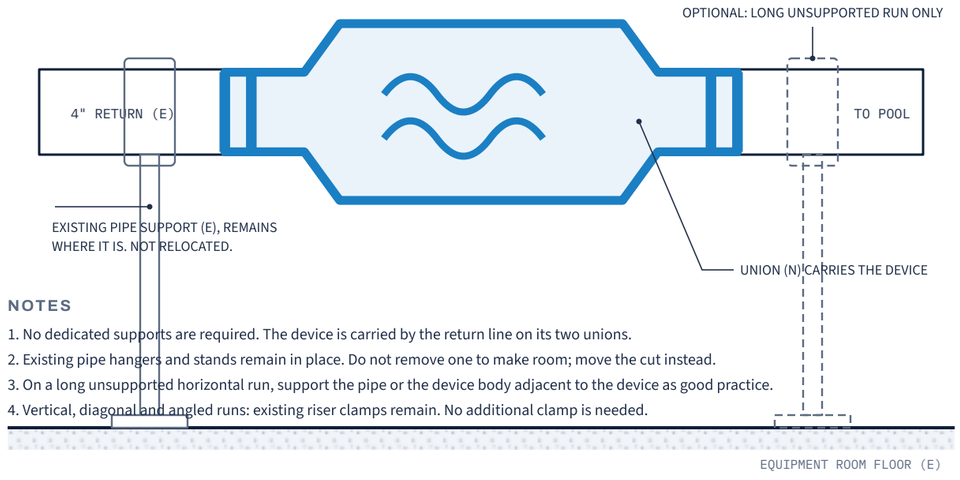
3" = 1'-0" · 4" SHOWN, 2", 3" AND 6" SIMILAR · JOINT PER ASTM D2855




WEEP AT A UNION UNDER PRESSURE: DEPRESSURIZE, LOOSEN, CLEAN FACES, RESEAT O-RING, RETIGHTEN BY HAND

## 4 M-301 PIPE SUPPORT AT DEVICE · HORIZONTAL RUN

NO SCALE · NO DEDICATED SUPPORTS REQUIRED · EXISTING SUPPORTS REMAIN




### NOTES

1. No dedicated supports are required. The device is carried by the return line on its two unions.
2. Existing pipe hangers and stands remain in place. Do not remove one to make room; move the cut instead.
3. On a long unsupported horizontal run, support the pipe or the device body adjacent to the device as good practice.
4. Vertical, diagonal and angled runs: existing riser clamps remain. No additional clamp is needed.
5. Hardware, where used: galvanized or fiberglass strut, PVC-coated or stainless clamps in chemical rooms. Not supplied with the device.
6. Winter: the device drains through the unions with the rest of the loop. No separate drain point is needed.

**AQUAREV WATER**

COMMERCIAL POOL SYSTEMS

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PROJECT

**Sample Commercial Pool Pump Room**

AquaRev water conditioning device installation

STATUS

Reference set. Not site-specific.

FOR

Engineers new to the device

SHEET TITLE

**Device & Connection Details**

| | |
|---|---|
| DRAWN | AquaRev Water |
| BASIS | Sample property |
| DATE | 22 SEP 2026 |
| SCALE | As noted |
| SIZE | ANSI B 17 × 11 |
| SET | 100 · REV A |

REVISIONS

| REV | DATE | DESCRIPTION |
|---|---|---|
| A | 09-22-26 | Reference issue |

Dimensions are from the Product Specification Guide 2026. Measure the unit in hand before marking pipe. Union and cement maker's instructions govern the joint.

REFERENCE SET. NOT A SITE-SPECIFIC DRAWING.

SHEET

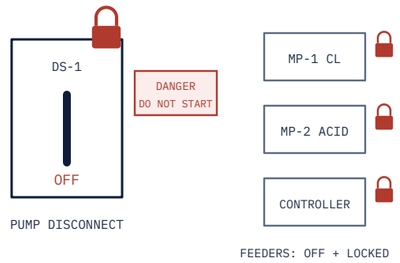
**M-301**

SHEET 5 OF 8

INSTALLATION SEQUENCE · PER LOOP · STEPS 01 TO 08 IN ORDER · STEP 01 IS NEVER SKIPPED

## 01 LOCK OUT AND TAG

PUMP + EVERY FEEDER ON THE LOOP

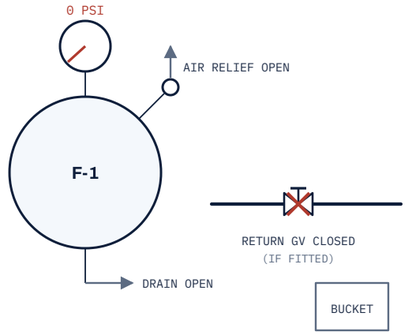



STOPPING FLOW WITH FEEDERS LIVE CAN CONCENTRATE CHEMICAL IN THE LINE

Switch off and lock the pump at its disconnect, and every chemical feeder on the loop: chlorinator or salt cell, acid or $CO_2$ feed, and the controller's dosing outputs. Tag each lock. Before this, photograph the carton label and record model and serial. Find IN and OUT and decide which end faces the pump.

## 02 ISOLATE AND DEPRESSURIZE

GAUGE MUST READ ZERO

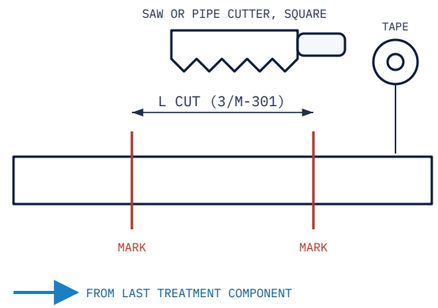



NO PRESSURE IN THE LINE BEFORE ANY CUT

Close the return-line isolation valves where fitted. Otherwise open the filter drain and let the line empty. Open the filter air relief and confirm the gauge reads zero. Put a bucket under the cut zone; the return leg still holds water. Open the door or run a fan: solvent cement in a closed pump room needs ventilation.

## 03 MEASURE, MARK AND CUT

SOCKET BOTTOM TO SOCKET BOTTOM

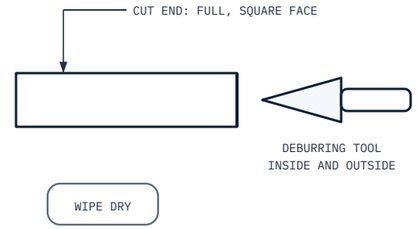



MEASURE TWICE. CUT ONCE.

Dry-fit both union tails to the device and measure socket bottom to socket bottom. That is the cut length; write it on the pipe. Mark on the return run after the last treatment component, as close to the pool return as the room allows, with 4 in. clear each side. Cut square at both marks and remove the section. Never cut to L1.

## 04 DEBUR AND CLEAN

SQUARE · CLEAN · DRY

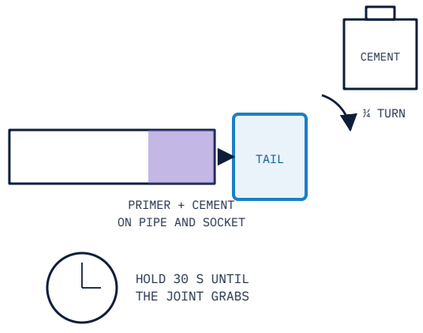



TAIL DRY-FITS 1/3 TO 2/3 OF SOCKET DEPTH BY HAND: PIPE IS IN TOLERANCE

Remove burrs inside and out on both cut ends. Wipe the ends and the union sockets clean and dry: no water, no dust, no old cement. Dry-fit a tail on each end; it should enter one third to two thirds of the socket depth by hand. Confirm all supplied parts are present: two union halves, two O-rings, the device.

## 05 PRIME AND CEMENT THE TAILS

FULL DEPTH · ¼ TURN · HOLD 30 S

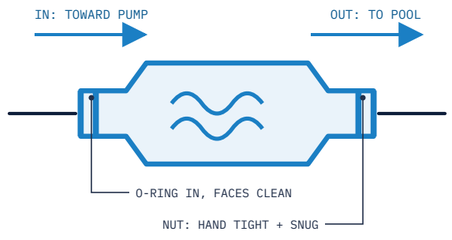



NUTS ORIENTED TO TURN BY HAND LATER

Slide the union nut onto the pipe first, facing the cut. Prime the pipe end and the tail socket, cement both, push the tail fully home with a quarter turn and hold 30 seconds. Wipe the bead. Repeat on the other end. Check the shelf date on primer and cement; use heavy-body cement on 4 in. and larger. Gloves, eye protection, ventilation.

## 06 SET THE DEVICE

IN TOWARD PUMP · HAND TIGHT + SNUG




NO WRENCH. PVC UNIONS SEAL ON THE O-RING, NOT ON TORQUE.

Check the IN arrow points toward the pump. Seat the device into the union tails with the O-rings in place and the faces clean. Confirm full insertion, then tighten each nut hand-tight plus a small snug. No wrench, no dope, no cement on the threads. No dedicated supports are needed; existing pipe supports stay where they are (4/M-301).

## 07 CURE, THEN REFILL

PER CEMENT MAKER · WHEN IN DOUBT, LONGER



TYPICAL CURE BEFORE PRESSURIZING

| PIPE | ROOM TEMP |
|---|---|
| 2" | 1 to 2 h |
| 3" and 4" | 2 to 4 h |
| 6" | 4 h or more |

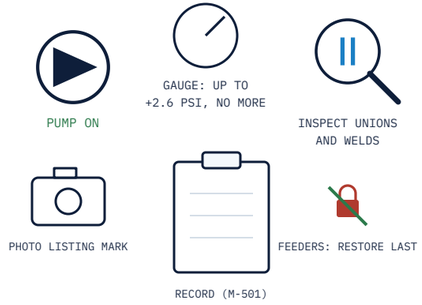
COOL OR DAMP ROOM: LONGER. MANUFACTURER'S TABLE GOVERNS.

MEANWHILE: CLOSE DRAINS, REFILL FILTER, LEAVE AIR RELIEF OPEN

Wait the cement manufacturer's full cure time for the pipe size and room temperature before any pressure. Two-inch joints at room temperature typically want one to two hours; larger sizes and cool, damp rooms want longer. Use the time to close drains, refill the filter and tidy the cut zone. Do not start the pump early to "test".

## 08 RESTART, CHECK AND RECORD

PUMP FIRST. FEEDERS LAST.




Open isolation valves. Remove the pump lockout only. Start the pump with the air relief open until water runs clear. Inspect both unions and both welds under pressure; a weep means reseat, not tighten harder. Read the filter gauge. Then restore feeders and controller. Photograph the install with the listing mark visible. Recheck for leaks after 24 h.

**AQUAREV WATER**

COMMERCIAL POOL SYSTEMS

aquarevwater.us

PROJECT

**Sample Commercial Pool Pump Room**

AquaRev water conditioning device installation

STATUS

Reference set. Not site-specific.

FOR

Engineers new to the device

SHEET TITLE

**Installation Sequence**

| | |
|---|---|
| DRAWN | AquaRev Water |
| BASIS | Sample property |
| DATE | 22 SEP 2026 |
| SCALE | None |
| SIZE | ANSI B 17 × 11 |
| SET | 100 · REV A |

REVISIONS

| REV | DATE | DESCRIPTION |
|---|---|---|
| A | 09-22-26 | Reference issue |

One loop, one device, one crew. Three to five hours including cement cure. Expanded from the Product Specification Guide and Implementation Guide §8.

REFERENCE SET. NOT A SITE-SPECIFIC DRAWING.

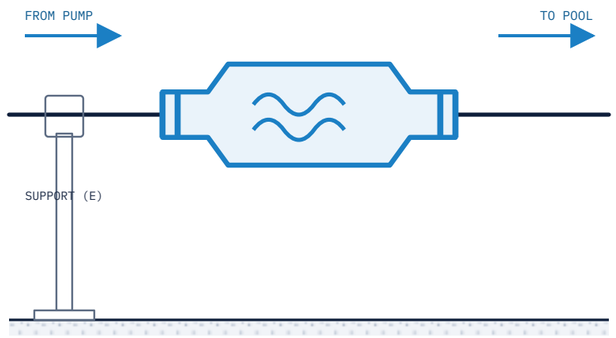
SHEET

**M-302**

SHEET 6 OF 8

## A M-401 HORIZONTAL RUN · PREFERRED

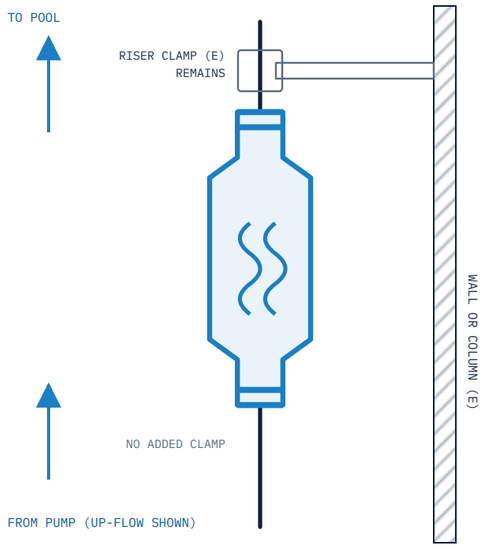
EASIEST TO INSPECT AND REMOVE




No dedicated supports. Existing hangers remain. Label faces out. Room above the body to lift it out after loosening both nuts. No other requirements.

## B M-401 VERTICAL RUN · EITHER DIRECTION

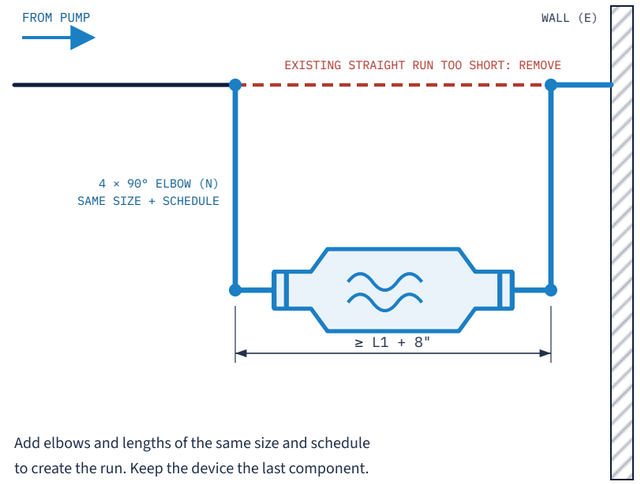
GRAVITY HAS NO EFFECT. IN ARROW STILL FACES THE PUMP.




Down-flow is equally acceptable. Keep hand access to both nuts.

## C M-401 OFFSET REROUTE · SHORT RETURN RUN

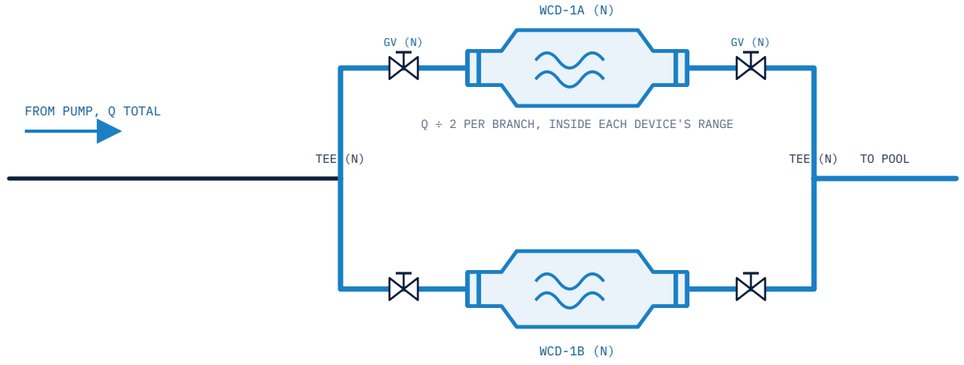
WHEN THE STRAIGHT RUN IS LESS THAN L1 + 8"




Add elbows and lengths of the same size and schedule to create the run. Keep the device the last component. Support the new run at its elbows as for any rerouted pipe. Elbows add friction; four 90s on 4 in. pipe add well under 1 psi at 417 gpm. Confirm pump margin.

## D M-401 PARALLEL DEVICES · FLOW ABOVE ONE MODEL'S RANGE

SYMMETRICAL BRANCHES, SAME SIZE, SAME LENGTH · CONSULT AQUAREV BEFORE ORDERING

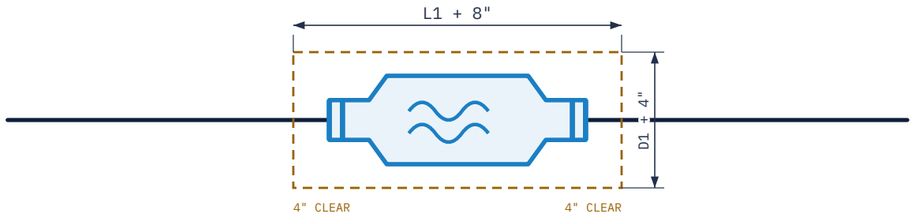



First choice is always the single model that matches the return pipe. Parallel devices are for loops whose flow exceeds that model's maximum. Branches are identical so the flow splits evenly; each device must see flow inside its range. Isolation valves let one branch be serviced with the pool running. Head loss stays 2.6 psi.

EXAMPLE: 1,400 GPM ON AN 8" RETURN, 8" UNIT NOT SELECTED → 2 × KD6X12X6 AT 700 GPM EACH (RANGE 440–1,100). REFER TO AQUAREV.

## E M-401 CLEARANCE ENVELOPE · ALL MODELS

FREE OF FITTINGS, VALVES, WALLS AND EQUIPMENT · HAND ACCESS TO BOTH NUTS




| MODEL | L1 | L1 + 8" RUN | D1 | D1 + 4" CLEAR | SHIP LB |
|---|---|---|---|---|---|
| KD2X4X2 · 2" | 20.25" | 28.25" | 5.00" | 9.00" | 8 |
| KD3X6X3 · 3" | 24.00" | 32.00" | 7.25" | 11.25" | 20 |
| KD4X8X4 · 4" | 28.19" | 36.19" | 9.34" | 13.34" | 35 |
| KD6X12X6 · 6" | 40.00" | 48.00" | 13.75" | 17.75" | 50 |

The device is removed for inspection by loosening both nuts and lifting it out, so keep D1 + 4 in. clear on the lift-out side. A 6 in. unit weighs 50 lb: plan two people or a lift point. Where the envelope cannot be met on the existing run, use arrangement C. Where flow exceeds the range, use arrangement D.

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FOR

Engineers new to the device

SHEET TITLE

**Orientation & Alternate Arrangements**

| | |
|---|---|
| DRAWN | AquaRev Water |
| BASIS | Sample property |
| DATE | 22 SEP 2026 |
| SCALE | None |
| SIZE | ANSI B 17 × 11 |
| SET | 100 · REV A |

REVISIONS

| REV | DATE | DESCRIPTION |
|---|---|---|
| A | 09-22-26 | Reference issue |

The device works in any orientation. These arrangements solve space and flow-range problems without moving any existing equipment.

REFERENCE SET. NOT A SITE-SPECIFIC DRAWING.

SHEET

**M-401**

SHEET 7 OF 8

## MATERIALS PER DEVICE

| ITEM | QTY | BY | SPECIFICATION |
|---|---|---|---|
| AquaRev water conditioning device, model per site assessment record | 1 | AquaRev | NSF/ANSI/CAN 50-2024, IAPMO R&T W-12547, PVC, lead-free |
| Schedule 40 PVC union halves with O-rings | 2 | AquaRev | Supplied in the carton with the device; verify present |
| PVC primer | 1 | Contractor | ASTM F656, purple, in date |
| PVC solvent cement | 1 | Contractor | ASTM D2564; heavy body for 4 in. and larger; in date |
| Spare pipe, same size and schedule as the return | 10 ft | Contractor | ASTM D1785 Sch 40 PVC, for reroute or recovery |
| Couplings and 90° elbows, same size and schedule | as req. | Contractor | ASTM D2466 Sch 40 PVC socket fittings |
| Pipe supports | as req. | Contractor | None required for the device; only where a rerouted run needs them |
| Lock-out devices and tags | per loop | Contractor | Pump disconnect plus every chemical feeder on the loop |
| Flow-direction label on pipe at device | 1 | Contractor | Optional; arrow toward pool, "AquaRev WCD, remove at unions" |

MODEL: BY RETURN PIPE SIZE, THEN FLOW INSIDE THE RANGE (1/M-301). CLEARANCE: E/M-401. FLOW WITHIN 10% OF A LIMIT, POOLS OVER 500,000 GAL, OR 8" AND LARGER RETURNS: REFER

## TOOLS AND SAFETY

Pipe cutter or fine-tooth saw · deburring tool · tape measure · marker · primer and cement daubers · clean rags · bucket

Gloves and eye protection · fan or open door for ventilation · no heat or open flame near primer and cement

Camera (carton label, cut zone with tape in frame, finished install with listing mark) · copy of the AHJ determination on site

## CLOSEOUT: KEEP IN THE PROJECT FILE

1. Site assessment record for each body of water, with photographs (Implementation Guide §3).
2. AHJ determination record with the written reply, or the permit and inspection sign-off (Implementation Guide §6–7).
3. IAPMO R&T listing, Product Specification Guide and warranty terms as issued at purchase.
4. The installation record, completed, with serial numbers, installer licence and photographs.

## INSTALLATION RECORD · TEMPLATE, ONE PER DEVICE

| | |
|---|---|
| PROPERTY / BODY OF WATER | |
| DEVICE TAG / MODEL | |
| SERIAL NUMBER | |
| RETURN PIPE OD / MATERIAL | |
| FLOW (GPM), HOW MEASURED | |
| CUT LENGTH USED (IN.) | |
| CEMENT TYPE / CURE ALLOWED | |
| FILTER GAUGE BEFORE / AFTER | _____ / _____ PSI |
| LEAK CHECK AT START-UP | □ DRY □ WEEP, CORRECTED |
| LEAK CHECK AFTER 24 H | □ DRY □ WEEP, CORRECTED |
| PHOTOS (LABEL, CUT, FINAL) | □ LABEL □ CUT □ FINAL |
| AHJ DETERMINATION ON SITE | □ YES □ PERMIT NO. _____ |
| INSTALLER / LICENCE NO. | |
| SIGNATURE / DATE | |

## CONTACTS

Sales and AHJ package support
water@aquarevwater.us · (832) 979-6758

Technical support and warranty claims
aquarev@hdcproductsinc.com · (808) 628-0208

Listing verification
pld.iapmo.org · File No. W-12547

Documents
aquarevwater.us/data · aquarevwater.us/warranty

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FOR
Engineers new to the device

SHEET TITLE
**Materials & Records**

| | |
|---|---|
| DRAWN | AquaRev Water |
| BASIS | Sample property |
| DATE | 22 SEP 2026 |
| SCALE | None |
| SIZE | ANSI B 17 × 11 |
| SET | 100 · REV A |

REVISIONS

| REV | DATE | DESCRIPTION |
|---|---|---|
| A | 09-22-26 | Reference issue |

Model is selected from the site assessment record (1/M-301 for ranges and dimensions). The record form is a template for any installation.

REFERENCE SET. NOT A SITE-SPECIFIC DRAWING.

SHEET
**M-501**
SHEET 8 OF 8